DATA VISUALIZATION AND STORYTELLING-Interview Questions and Answers

1. What is the importance of data visualization?

   Data visualization is important because it converts complex data into charts, graphs, and dashboards that are easy to understand. It helps people identify patterns, trends, relationships, and unusual values quickly. It also supports faster and better business decisions. For example, a sales manager can use a monthly sales chart to identify the best-performing products and regions.
2. When do you use a pie chart vs bar chart?

I use a pie chart when I need to show how a small number of categories contribute to a whole, such as the percentage of total sales by product category. I use a bar chart when I need to compare values across different categories, such as sales by region or product. Bar charts are usually better when there are many categories or when accurate comparisons are important.

3. How do you make visualizations more engaging?

   I make visualizations engaging by choosing the correct chart, using clear titles and labels, and keeping the design simple. I use a consistent and limited colour palette, highlight important values, and add filters or tooltips when useful. I also include a short explanation of the main finding so that viewers understand why the chart matters.
4. What is data storytelling?

   Data storytelling is the process of combining data, visualizations, and a clear explanation to communicate an insight or message. It does not just show numbers; it explains what happened, why it may have happened, and what action could be taken. For example, a report may show that sales decreased in one region, explain the possible reasons, and recommend a focused marketing campaign.
5. How do you avoid misleading visualizations?

   I avoid misleading visualizations by using an appropriate chart type and a clearly labelled scale. I do not distort axes, hide important data, use 3D effects unnecessarily, or choose colours that create a false impression. I provide the correct units, time period, and data source, and I check that the chart represents the data accurately.
6. What are best practices in dashboard design?

A good dashboard should have a clear purpose and be designed for its intended users. I place the most important KPIs at the top, group related charts together, and use consistent colours, fonts, and formats. I avoid clutter, provide meaningful titles, use filters only when needed, and make sure the dashboard is easy to read. I also validate the figures before sharing it.

7. What tools have you used for visualization?

I have learned to use Power BI and Tableau for creating visual reports and dashboards. In Power BI, I can work with data, create charts, use filters, and build interactive dashboards. Tableau can also be used to explore data and create interactive visualizations. I am continuing to improve my skills by practising with sales datasets such as the Superstore dataset.